# Elliot Kaiser

206-707-2649 | elliotk1@outlook.com | linkedin.com/in/elliot-kaiser | elliotkaiser.me | USA & CAN Citizen

## Education

**University of Toronto** — Toronto, ON
*BASc, Mechanical Engineering + PEY Co-op (GPA: 3.9/4.0, Dean's Honours List)* — *Sep 2025 – May 2029*
- Varsity Men's Field Lacrosse (15+ hrs/week)

## Experience

**Undergraduate Research Assistant – Systems & Hardware** — Oct 2026 – Present
*UofT Lin Lab, Department of Cell & Systems Biology* — *Toronto, ON*
- Reducing end-to-end latency (~90 ms baseline) of a closed-loop 360° zebrafish VR system, building Arduino photodiode and 240 fps timing rigs (C++/Python) to profile the camera, Godot, and projector pipeline

**Undergraduate Research Assistant – Composites** — Sep 2026 – Present
*UofT Multifunctional Composites Manufacturing Lab (MCML)* — *Toronto, ON*
- Deconvolve XPS carbon spectra in OriginPro to quantify the surface chemical bonding states of carbon fiber
- Perform a wet-layup procedure for nanoparticle-reinforced carbon fiber/epoxy laminates, using pulsed sonication and vacuum degassing to disperse particles and reduce voids

**Mechanical Engineering Intern** — May 2026 – Aug 2026
*Renton Coil Spring* — *Renton, WA*
- Designed and built a spring failure detection system (piezo sensor, ESP32, safety relay), enabling unattended 24/7 operation and preventing 300,000 wasted cycles per failed spring
- Designed and 3D printed universal SOLIDWORKS fixtures for non-standard spring geometries on a Rockwell hardness tester, achieving repeatable measurements within ±1 HRC
- Fabricated 10+ aerospace-grade springs on 2D and 3D CNC machines and ran side-by-side quality comparisons
- Converted 15+ legacy 2D drawings into 3D SOLIDWORKS models, reducing supplier lead times

## Design Teams

**Mechanical Engineer** — Sep 2026 – Present
*Autonomous Rover Team - University of Toronto Robotics Association* — *Toronto, ON*
- Derive first-principles load cases (traction, incline, acceleration) from competition rules to size motors, wheels, bearings, and drive type for a 15–40 kg 2WD rover within budget
- Iterate a custom-machined drive shaft with ANSYS Mechanical FEA across all load cases, raising the minimum factor of safety to 2.4 (target 1.5)
- Design and trade off drivetrain layouts in SOLIDWORKS against packaging, cost, and serviceability

**Hull and Structural Design Engineer** — Sep 2025 – May 2026
*UofT Concrete Canoe* — *Toronto, ON*
- Rewrote in-house hull performance simulation from C++ to Python, modeling 30%+ more race parameters
- Converted generated hull meshes into SOLIDWORKS CAD and designed gunwales, handles, and seats

## Projects

**Smart Scale** | *SOLIDWORKS, OrcaSlicer, Python, Flask, PostgreSQL, Raspberry Pi* — Jul 2026 – Aug 2026
- Integrated a 5 kg cantilever load cell, HX711 ADC, and Raspberry Pi 4 in a custom 3D-printed enclosure, implementing median filtering and real-time calibration for ±1 g accuracy
- Developed a full-stack Supabase (PostgreSQL) web app for guided recipes, meal logging, and nutrition goals

**Volunteer Management Software** | *Next.js, React, TypeScript, PostgreSQL* — Jan 2026 – Apr 2026
- Built a mobile app centralizing volunteer coordination for a 120-member sailing club, cutting admin overhead 50%+

## Technical Skills

**Mechanical Design**: SOLIDWORKS (CSWA), ANSYS Mechanical (FEA), OnShape, AutoCAD, GD&T, DFM/A
**Embedded Systems**: ESP32, Arduino, Raspberry Pi, Sensor Integration, KiCad, Soldering
**Software**: Python, C/C++, MATLAB, OriginPro, Godot, Java, JavaScript/React, Flask, PostgreSQL, Microsoft Office
**Manufacturing**: CNC Machining, Composite Wet Layup, Vacuum Degassing, 3D Printing (FDM/OrcaSlicer), Shop Tools